

# Robotics and Automation: The Role of Artificial Intelligence in Autonomous Robots and PLC-Based Automation Systems

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**Course:** Robotics and Automation

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## 1. Introduction

Robotics and automation have become essential technologies in modern industries. They improve productivity, increase accuracy, reduce operational costs, and enhance workplace safety. Recent developments in Artificial Intelligence (AI), sensors, communication networks, and control systems have enabled robots to perform increasingly complex tasks with minimal human intervention.

Automation systems are widely used in manufacturing, transportation, healthcare, agriculture, logistics, and smart homes. Autonomous robots can sense their environment, make decisions, and execute tasks independently. At the same time, Programmable Logic Controllers (PLCs) remain the backbone of industrial automation due to their reliability, flexibility, and real-time control capabilities.

This mini-project explores the role of AI in robotics and automation, autonomous robotic systems, PLC-based automation, and presents a conceptual design of a robotic pick-and-place unit.

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## 2. Objectives

The main objectives of this project are:

1. To study the role of Artificial Intelligence in robotics and automation.
2. To understand the working principles of autonomous robots.
3. To analyze PLC-based automation systems used in industry.

4. To develop a conceptual robotic unit design.
  5. To demonstrate a simple robotic pick-and-place operation.
  6. To evaluate the advantages and challenges of intelligent automation systems.
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### **3. Role of Artificial Intelligence in Robotics and Automation**

Artificial Intelligence enables machines to perform tasks that normally require human intelligence. AI allows robots to learn from data, recognize patterns, make decisions, and adapt to changing environments.

Key AI technologies used in robotics include:

#### **Machine Learning**

Machine learning enables robots to improve their performance through experience. The robot can analyze data from sensors and optimize its behavior over time.

#### **Computer Vision**

Computer vision allows robots to identify objects, detect obstacles, read labels, and recognize human gestures through cameras and image-processing algorithms.

#### **Path Planning**

AI algorithms help robots determine the safest and most efficient route to reach a destination while avoiding obstacles.

#### **Decision Making**

Intelligent robots can evaluate multiple situations and choose appropriate actions without direct human control.

Applications of AI in robotics include:

- Autonomous vehicles
- Delivery drones
- Industrial robots
- Warehouse automation
- Medical robots
- Smart manufacturing systems

The integration of AI significantly increases robotic efficiency, flexibility, and autonomy.

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#### 4. Autonomous Robots

An autonomous robot is a robotic system capable of sensing its environment, processing information, making decisions, and performing tasks without continuous human intervention.

##### Components of Autonomous Robots

###### Sensors

Sensors collect information about the environment.

Examples:

- Cameras
- Ultrasonic sensors
- LiDAR
- Infrared sensors
- GPS modules

###### Controller

The controller acts as the robot's brain and processes sensor information.

Examples:

- Microcontrollers
- Industrial computers
- Embedded processors

###### Actuators

Actuators convert electrical signals into physical movement.

Examples:

- Servo motors
- Stepper motors
- DC motors

## Artificial Intelligence Module

AI software analyzes information and determines the robot's actions.

### Working Principle

The robot follows a continuous cycle:

1. Sense environment.
2. Process information.
3. Make decisions.
4. Execute actions.
5. Receive feedback.

This cycle allows robots to perform tasks independently and adapt to changing conditions.

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## 5. PLC-Based Automation System

A Programmable Logic Controller (PLC) is an industrial digital computer used to automate electromechanical processes.

PLCs are commonly used because they are:

- Reliable
- Durable
- Easy to program
- Suitable for industrial environments

### Basic PLC System Components

#### Inputs

Input devices provide information to the PLC.

Examples:

- Push buttons
- Sensors
- Limit switches

- Proximity sensors

### PLC Controller

The PLC processes input signals according to a programmed logic sequence.

### Outputs

Output devices perform actions based on PLC commands.

Examples:

- Motors
- Relays
- Valves
- Indicators

### PLC Operation Sequence

1. Read inputs.
2. Execute program logic.
3. Update outputs.
4. Repeat continuously.

### Industrial Applications

- Conveyor systems
- Packaging machines
- Robotic assembly lines
- Traffic control systems
- Smart manufacturing

PLCs remain a critical component of Industry 4.0 automation systems.

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## 6. Robotic Unit Design Concept

### Proposed System

The proposed design is a **4-DOF Pick-and-Place Robotic Arm** controlled through a PLC or microcontroller.

## Purpose

The robot is designed to:

- Pick an object from Location A.
- Move the object.
- Place it at Location B.
- Repeat automatically.

## Main Components

### Base

Provides stable support and rotational movement.

### Shoulder Joint

Moves the arm vertically.

### Elbow Joint

Extends the reach of the robot.

### Wrist Joint

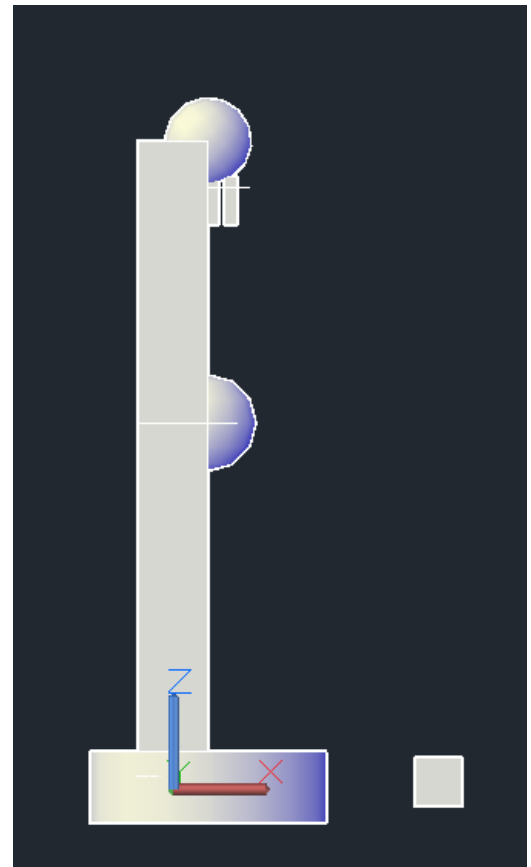
Adjusts the orientation of the gripper.

### Gripper

Holds and releases objects.

### Degrees of Freedom (DOF)

| Joint    | Motion   | DOF |
|----------|----------|-----|
| Base     | Rotation | 1   |
| Shoulder | Rotation | 1   |
| Elbow    | Rotation | 1   |
| Wrist    | Rotation | 1   |



Total Degrees of Freedom = 4

### System Inputs

- Object detection sensor
- Position sensor
- Start/Stop push button

### System Outputs

- Servo motors
- Gripper actuator
- Status indicator LEDs

## 7. Automation Flowchart

### System sequence:

START

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Detect Object

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Object Present?

→ No → Wait

→ Yes

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Move Arm to Pick Position

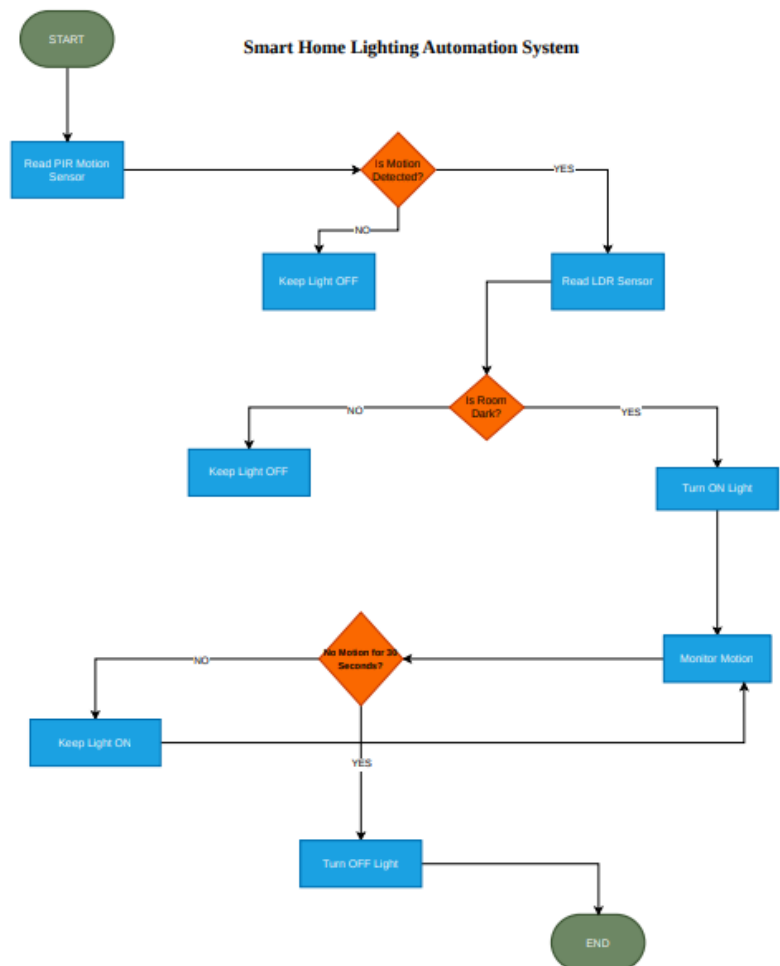
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Close Gripper

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Lift Object

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Move to Place Position



Open Gripper



Return to Home Position



Repeat Process

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## 8. Expected Results

The proposed robotic system is expected to:

- Automatically detect and move objects.
  - Reduce manual labor requirements.
  - Improve operational efficiency.
  - Demonstrate the integration of AI and automation principles.
  - Provide a practical example of industrial robotic applications.
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## 9. Conclusion

Robotics and automation continue to transform modern industries through intelligent control systems and autonomous operation. Artificial Intelligence enables robots to perceive, learn, and make decisions, while PLCs provide reliable industrial control. The proposed 4-DOF robotic arm demonstrates the practical integration of robotics, automation, sensing, and control technologies. Such systems contribute to improved productivity, enhanced safety, and greater operational efficiency, making them essential components of future smart factories and automated environments.

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